A PROJECT REPORT

ON

"Canteen Management System"

By

Ananya Tiwari

Roll No. 23081010294

A project report submitted in partial fulfillment of the requirements for the Degree of B.Sc in Computer Science of to Ramwati Raj Bahadur Degree College PrithvipurEtawah .

Submitted By	Submitted To
Ananya Tiwari	Dr. Saurabh Gupta

STUDENT DECLARATION

I hereby declare that the dissertation project report entitled

"Canteen Management System"

Submitted in partial fulfillment of the requirements for the Degree of
B.Sc in Computer Science

To Ramwati Raj Bahadur Degree College Prithvipur Etawah, it is my original work and not submitted for the award of any other Degree, Diploma, Fellowship, or any other title or prizes .

Place : Etawah **Student Name: Ananya Tiwari**

Date : **Roll No. 23081010294**

CERTIFICATE

This is to certified that project report entitled

"Canteen Management System"

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE DEGREE OF

DEGREE OF
B.Sc. IN COMPUTER SCIENCE
Of

Ramwati Raj Bahadur Degree College PrithvipurEtawah,

Ananya Tiwari

Has worked under my supervision and guidance and that no part of this report has been submitted for the award of any other , degree, diploma, fellowship or any other similar title or prizes and that work has not been published in any journal or magazine.

Roll No. 23081010294 **CERTIFIED**

ACKNOWLEDGEMENT

The successful accomplishment of any task is incomplete without acknowledging the contributing personalities who both assisted and inspired and lead us to visualize the things that turn them into successful stories for our successors.

First of all I thank the almighty God for his grace bestowed on us throughout this project.s

I am very pleased to present this project report at Zelore Technologies. Pvt.Ltd. My project in this organization has been a prolific experience and excellent source of gaining knowledge I would like to express my thanks to Mr.AnuragUpadhya . HR Head of Visions Wave Technologies. Pvt.Ltd and I would also like to thank the other member of the staff who had taken their time off to help me in study again I am thankful to whole HR Department for their co-operation during my project work.

My special thanks to my Dr. SaurabhGupta, Vishar Nagar who guide me with the timely advice and expertise and have helped remarkably to complete the project .

TABLE OF CONTENTS

1. **Acknowledgement**
2. **Certificate of the Project Guide**
3. **synopsis of the project**
4. **Organization Profile**
5. **Problem Formulation**

System Objective

System purpose

System Context

Need Recognition

6. **System Study**

System Planning

7. **System Analysis**

Process Flow Analysis

Accommodation

a. Work Flow
b. Data Flow Diagram(DFD)
c. Entity Relationship diagram(ERD)

9.System Design

1. Table Structure
2. User Interface (Form Design)
3. Report Design
4. Hardware and Software Specification

10. System Testing

1.Introduction

2. Unit Testing

3. Integration Testing

4. System Testing

5. Acceptance Testing

Introduction

Many organizations provide Canteen as an additional subsidized facility to their employees as a welfare measure. The usage of the canteen by the employees depends on many factors like centralized food distribution center with swift and simplicity in operations. Exultsoft CMS design provides a user friendly system that facilitates quick and efficient operations to cover larger section of employees within a specified time. **CMS is paperless and has almost cashlesstransactions.** This system is also compatible in an environment where multiple caterers operate in a closed environment or at multiple locations.

FEATURES

1. CMS can be implemented in an organization where the employees have cards (any type of identification cards) for unique identification.
3. CMS can be integrated with an existing (or new), Personnel Information system, Time and Attendance System, and Access Control System etc.
4. CMS can be integrated with different POS terminals depending on the Customer needs.
5. CMS allow user to keep record paperless.
6. Manual entry of transactions is feasible.
7. Daily menu creation by the caterer is possible using CMS.
8. Ease in selection from menu by Caterers.

BENEFITS

- CMS is independent of the technology used for identification of employees.
- Provides cashless and paperless transactions.
- Provides a choice to the employee on the mode of operation (debit/credit).
- Accounting is made easier between the company and the caterers.
- Reconciliation between caterers is made easy.
- Provides salary-debit facility for employees in credit scheme, for canteen facilities used.
- Provides a fast and efficient service.
- Use of SodexHo coupons instead of cash.
- **Reduces wastage**.

Tools/Platforms and Languages Used

Hardware Configuration

Processor: 845 Intel Processor

Main Memory: 128 MB RAM

HDD: 20 GB

FDD: 1.44 MB

SVGA Color monitor

Logitech mouse

105 keys keyboard

Software

Operating System: WINDOWS 98 SE

Programming Language: C++

Documentation: MS-WORD 2000

Backup media: Hard Disk Backup

Why using C++

Developments in software technology continue to be dynamic. New tools and techniques are announced in quick succession. This has forced the software engineers and industry to continuously look for new approaches to software design and development, and they are becoming more and more critical in view of the increasing complexity of software systems as well as the highly competitive nature of the industry.

These rapid advances appear to have created a situation of crisis within the industry. The following issues need to be addressed to face this crisis:

- How to represent real-life entities of problems in system design?
- How to design systems with open interfaces?
- How to ensure reusability and extensibility of modules?
- How to develop modules that are tolerant to any changes in future?
- How to improve software productivity and decrease software cost?
- How to improve the quality of service?
- How to manage time schedules?
- How to industrialize the software development process?

Many software products are either not finished, or not used, or else are delivered with major errors.

To reduce such errors software technology evolution gradually increases. Since,

The invention of the computer, many programming approaches has been tried. These include techniques such as modular programming and structured programming.

Procedure-Oriented Programming

Conventional programming, using high-level languages such as COBOL, FORTRAN and C, is commonly known as procedure oriented programming (POP).

Problems with POP

- In the procedure-oriented approach, the problem is viewed as a sequence of things to be done such as reading, calculating and printing.
- In a large program it is very difficult to identify what data is used by which function.
- Another serious drawback with the procedural approach is that it does not model real world problems very well. This is because functions are action oriented and do not really corresponds to the elements of the problem

Object–Oriented Programming

The major motivating factor in the invention of object-oriented approach is to remove some of the flaws encountered in the procedural approach. OOP treats data as a critical element in the program development and does not allow it to flow freely around the system. It ties data more closely to the functions that operate on it and protects it from accidental modification from outside functions.

Object – oriented programming is the most recent concept among programming paradigms. We define "object-oriented programming as an approach that provides a way of modularize problems by creating partitioned memory area for both data and functions that can be used as templates for creating copies of such modules on demand.

Characteristics of OOP languages

1. **Object**: Objects are entities, which can exist individually. It has its own properties and methods, where properties define the outlook of the object and methods define their procedures.

2. **Class**: It is a template used to define different objects of same type.

3. **Encapsulation**: The data and the methods, which operate on the data, are combined and placed in a group, this phenomenon is known as encapsulation, and the group is known as the object.

4. **Abstraction**: It means hiding of the data of one object of a class from another object of the same class.

5. **Inheritance**: Inheritance is the property by which an existing class can be used to create new classes, by deriving all the properties and methods of the old class to the new class and also adding new properties /methods in the new class. The old class is known as a base class or super class.
6. **Polymorphism**: Polymorphism means "One interface and multiple methods" i.e. one interface can be used to provide different functionalities.

There are two kinds of polymorphism:

a) **Compile time polymorphism**: It is also known as early binding, as the interfaces are bind with their methods at compile time. It is accomplished using function overloading and operator overloading.
b) **Run time polymorphism**: It is also known as late binding. In run time polymorphism the interface and its method bind at the time of execution. It is accomplished using virtual function.

7. Multiple Inheritance: When two or more classes are used to define a single class, then it is known as multiple inheritance.

Benefits of OOP

Object oriented contributes to solution of many problems associated with the development quality of software products. The new technology promises greater programmer productivity, better quality of software and lesser maintenance cost. The principal advantages are:

- Through inheritance, we can eliminate redundant code and extend the use of existing classes.
- We can build programs from the standard working modules that communicate with one another rather than having to start writing the code from scratch. This leads to saving of development time and higher productivity.
- The principle of data hiding helps the programmer to build secure programs that cannot be invaded by code in other parts of the program
- It is possible to have map objects in the problem domain to those in the program.
- It is easy to partition the work in a project based on objects.
- The data centered design approach enables us to capture more details of a model in implemental form,
- Object-oriented systems can be easily upgraded from small to large systems.
- Message passing techniques for communication between objects make the interface descriptions with external systems much simpler.

Software complexity can be easily managed.

Definition - What does C++ Programming Language mean?

C++ is an object oriented programming (OOP) language, developed by BjarneStroustrup, and is an extension of C language. It is therefore possible to code C++ in a "C style" or "object-oriented style." In certain scenarios, it can be coded in either way and is thus an effective example of a hybrid language.

C++ is a general purpose object oriented programming language. It is considered to be an intermediate level language, as it encapsulates both high and low level language features. Initially, the language was called 'C with classes' as it had all properties of C language with an additional concept of 'classes'. However, it was renamed to C++ in 1983.

It is pronounced "C-Plus-Plus."

Techopedia explains C++ Programming Language

C++ is one of the most popular languages primarily utilized with system/application software, drivers, client-server applications and embedded firmware.

The main highlight of C++ is a collection of pre-defined classes, which are data types that can be instantiated multiple times. The language also facilitates declaration of user defined classes. Classes can further accommodate member functions to implement specific functionality. Multiple objects of a particular class can be defined to implement the functions within the class. Objects can be defined as instances created at run time. These classes can also be inherited by other new classes which take in the public and protected functionalities by default.

C++ includes several operators such as comparison, arithmetic, bit manipulation, logical operators etc. One of the most attractive features of C++ is that it enables the overloading of certain operators such as addition.

A few of the essential concepts within C++ programming language include polymorphism, virtual and friend functions, templates, namespaces and pointers.

Project Planning

(1). The Concept

The concept behind software project planning is to provide a framework that enables the manager to make reasonable estimates of the resources, cost and schedule. The appropriate estimates are formulated within a time limit and the objective is achieved through a process of information discovery, which leads to reasonable estimates.

In Our Context

We studied and went through all set of tasks and activities regarding software and identified them before software project is to be implemented. With the help software project planning we constituted an identified works and resources required to complete the project and also the time required for the completion of project. Planning helped us in decision-making process while we were studying how to initiate and what steps to follow to accomplish the task. We also planned to finish the assigned task within appropriate cost, effort and resources to maintain the system feasible.

Our planning was about all kinds of resources required and estimation regarding total cost. We also finalized how much time it will take to launch the web site with in the given constraint of the cost, effort and resources. In resources we also considered human resources, reusable software resources and environmental resources.

(2). Software Scope:

Scope of software is the parameter under which the software is applicable or it is the unambiguous and understandable domain of the software in which the system is applicable. The scope of the software can be determined by following factors:

The Context: By understanding the context of the software (i.e. how does the software to be built to be fit into a large system, product, or business context and what constraint are required as a result of the context?), we can easily determine the scope of the software.

The Information objectives: Here, we try to know the user visible outputs and inputs. By understanding the end-user visible data objects (inputs/outputs), we determined the range of the information on which we have to work upon and for what purpose.

Function and performance: Here, we tried to know about the different functions, which transform the input data into output. By knowing about the different functions, we came to know about the different processes of the system.

The above said factors and their analysis helped us to exploit and implement the following features of software scope:

Reusability: Reusability is possible as and when we required in this application. We can update its next version. Reusable software reduces design, coding and testing cost by amortizing effort over several designs. Reducing the amount of code also simplified understanding, which increases the likelihood that the code is correct. We followed up both types of reusability as sharing of newly written code within a project and reuse of previously written code on new projects.

Extensibility: This application software is extended in ways that its original developers may not expect. The following principles enhance extensibility like Hiding data structures, avoiding traversing multiple links or methods, avoiding case statements on object type and distinguishing public and private operations.

Robustness: Its method is robust and it will not fail even if it receives improper parameters. There are some alert pages and messages are flashed out with some dialogue boxes to warn and inform the end user about the current processes going on. It also interacts with the user by alerting them about invalid parameters.

Understandability:A method is understandable if anyone other than the developer of the method can understand the code (as well as the developer after a time-span).

Cost-effectiveness: Its cost is under the budget and developed within given time period. It is always desirable to aim for a system with a minimum cost subject to the condition that it must satisfy all the requirements.

To implement Software Scope we considered it by looking out on the future scope of the software. The software is being implemented on a web site having server any were and Clint any computer on the net. The codes being embedded and installed at the root directory of the server with support for SQL and Internet Information Services. The database space taken at SQL server and connected with the site server will be main power for the resource utilization on the site. The software scope enlarges ultimately with the use of Internet technologies when universities organization institutes and individual can use and extract the vast and huge possibility of this site engulfed by the net technologies.

Portability: Since it is an Internet based application so its portability and usability depends upon the client connected with the internet. The interface designed that is the web page designing which is one of the major part of web application because it is the first impression regardless of the value of its contents interface should must grab a potential user immediately.

(3). Resource Estimation

The second task of planning is estimation of resources required to accomplish the software development effort. The development environment encapsulates hardware and software tools are the foundation of the resource and provide the infrastructure to support the development effort. At a higher level we encounter reusable software components – software building blocks that can dramatically reduce development costs and accurate delivery. At the highest level is the primary resource – people.

a). Human Resources: Human resource is the primary resource and lies at the topmost level of the resources hierarchy. It is the resource, which says about the people who are technically sound and develops the software. We started by evaluating scope and studying the skills required to complete the development. Both organization position and specialty are specified. Since we are two in our team with one project coordinator so we evaluated our skills and updated ourselves with the required skills and divided the work among us and so calculated the estimated schedule.

b). Reusable Software Resources: It lies at the second priority level of resources but any discussion would be incomplete without recognition of reusability. Since the task we are assigned is a new one and is to be developed from the very bottom level so we do not perform the planning and estimation regarding reusable software resources.

c). Environmental Resources: The environment that supports the software project often called a software engineering environment (**SEE**) incorporates hardware and software. Hardware provides a platform that supports the tools (software) required to produce the work products that are an outcome of good software engineering practices. Because most software organizations have multiple constituencies that require access to the **SEE**, a project planner must prescribe the time window required for hardware and software and verify that these resources will be available.

When a computer-based system (incorporating specialized hardware and software) is to be engineered, the software team may require access to hardware elements being developed by other engineering teams. The software project planner must specify each hardware element.

(4). <u>Project Estimation:</u> Software is most expansive element of any computer-based information system. So, the estimation of cost and effort applied to the project is crucial factor in system development. The major problem with software project estimation is that there are so many variables that affect the estimation and these too are difficult to quantify. Some of the variables that affect the estimation are – human, technical, environmental, and political. This does not mean we don't have the solution. There are a number of techniques, which has been developed for the estimation purpose. Some of the important techniques we adopted are as follows:

a) Decomposition Techniques
b) Empirical Estimation Models and
c) Automated Estimation Tools

(5). <u>Decomposition:</u> Software project estimation is a form of problem solving, and in most cases, developing a cost and effort estimate for a software project is too complex to be considered in one piece. So, we decompose the problem, and recharacterize it as a set of smaller problems. But no matter which technique we apply for our project, the first primary step in software project estimation is "**The determination of the size of the software project**

The accuracy of estimation of software project is dependent on the degree to which the size of the software has been estimated. The direct method to find the size of the software is to find the LOC (Lines of Code) of the software but this method is very difficult at the very beginning of the project. Because even the most experienced people will find it next to impossible to predict LOC before hand. There are various approaches to major the size of the software. Some of the important approaches are as follows:

a) Fuzzy logic Sizing
b) Function Point Sizing
c) Standard Component Sizing and
d) Change Sizing

Here, in our case, we have used the functional point sizing of the project. Let us have a brief review of the Function Point sizing concept.

Function Point Sizing: Albrecht first proposed the concept of functional point. Function Points are derived using an empirical relationship based on countable (direct) measures of software's information domain and assessment of software complexity. Functional points are computed by the formula

FP= Count Total X [0.65 + 0.01 X

Where, Fi is the value in the range of zero to five for various types of complexity factors in the project.

Software Engineering Paradigm

Introduction

Web – Engineering, as software engineering is a layered technology, which comprises of four independent layers as:

- **a quality focus layer,**
- **the process layer,**
- **the methods layer, and**
- **the tools layer.**

The foundation for the software engineering is the process layer and together with the technology layer helps the rational and timely development of computer software. The method layer provides a technical way to perform a broad array of tasks regarding requirements analysis, design, program construction, testing, and maintenance. The tools layer encompasses the software engineering tools to provide automated or semi automated support for the process and the methods.

In real world, the software development is a teamwork that incorporates a development strategy that encompasses the process, methods, and tools layers and so the strategy is termed as a process model or Software Engineering Paradigms

According to Fritz Bauer "Software engineering is the establishment and use of sound engineering principles in order to obtain economically software that is reliable and works efficiently on real machines". There are several models suggested to go through the software process. The one most suitable & we opted the linear sequential model to develop the software.

a). System Engineering: The software is always part of a large system and is built using modular approach. The very scratch work begins by establishing requirements for all system elements and then allocating some subset of these requirements for all system elements and then allocating some subset of these requirements to the software. This system view is essential whenever the software is intended to interact with other system elements such as hardware, people, and database. In due course the system engineering and the analysis encompass requirements gathering at the system level with a small amount of top-level design and analysis.

Information engineering encompasses requirements gathering at the strategic business level and at the business area level.

b). Software Requirement Analysis: In the very first approach of software development, the requirements gathering process is performed and later intensified and focused on the software. To understand the various characteristics of the software to be built, first we must have to understand the information domain for the software, and the required function, behavior, performance, and interface. Requirements for both the system and the software are documented and must be reviewed with the customer. Keeping this software engineering paradigm we are having a proper documentation on our web site so that Customer's requirement should be reviewed with the user. It helped us to minimize the distance between the actual requirement and the software required and hence kept and will keep the customer's satisfaction very high.

c). Design: The design is almost and always a multi step process and focuses on four distinct attributes of a program:

- **Data Structure**
- **Software Architecture**
- **Interface Representations and**
- **Procedural Detail**.

The design process translates requirements into a representation of the software that can be assessed for quality before coding begins. Like requirements, the design is documented and becomes part of the software configuration.

d). Code Generation: The design is translated into a machine-readable form using some programming tools.

e). Testing: Once the design is converted into machine-readable form the program testing phase starts. The testing process on our software mainly focused on the logical internals of the software, ensuring that all statements have been tested properly, and on the functional externals. Through testing we mainly intended to uncover errors and ensured that the defined input produces the desired results that agree with required results in the specification.

f). Maintenance: Software will undoubtedly undergo change after it is delivered to the customer. Change will occur due to either errors have been encountered or the software have been adapted to accommodate in its external environment or the customer requires functional or performance enhancements. Software support/ maintenance reapply each of the preceding phases to an existing program rather than a new one. The linear nature of the classic life cycle worked perfect without any "blocking states" as the project is to be developed by me under the guidance of the project coordinator only & hence we started the next phase as soon as the current phase is finished without waiting for any one else.

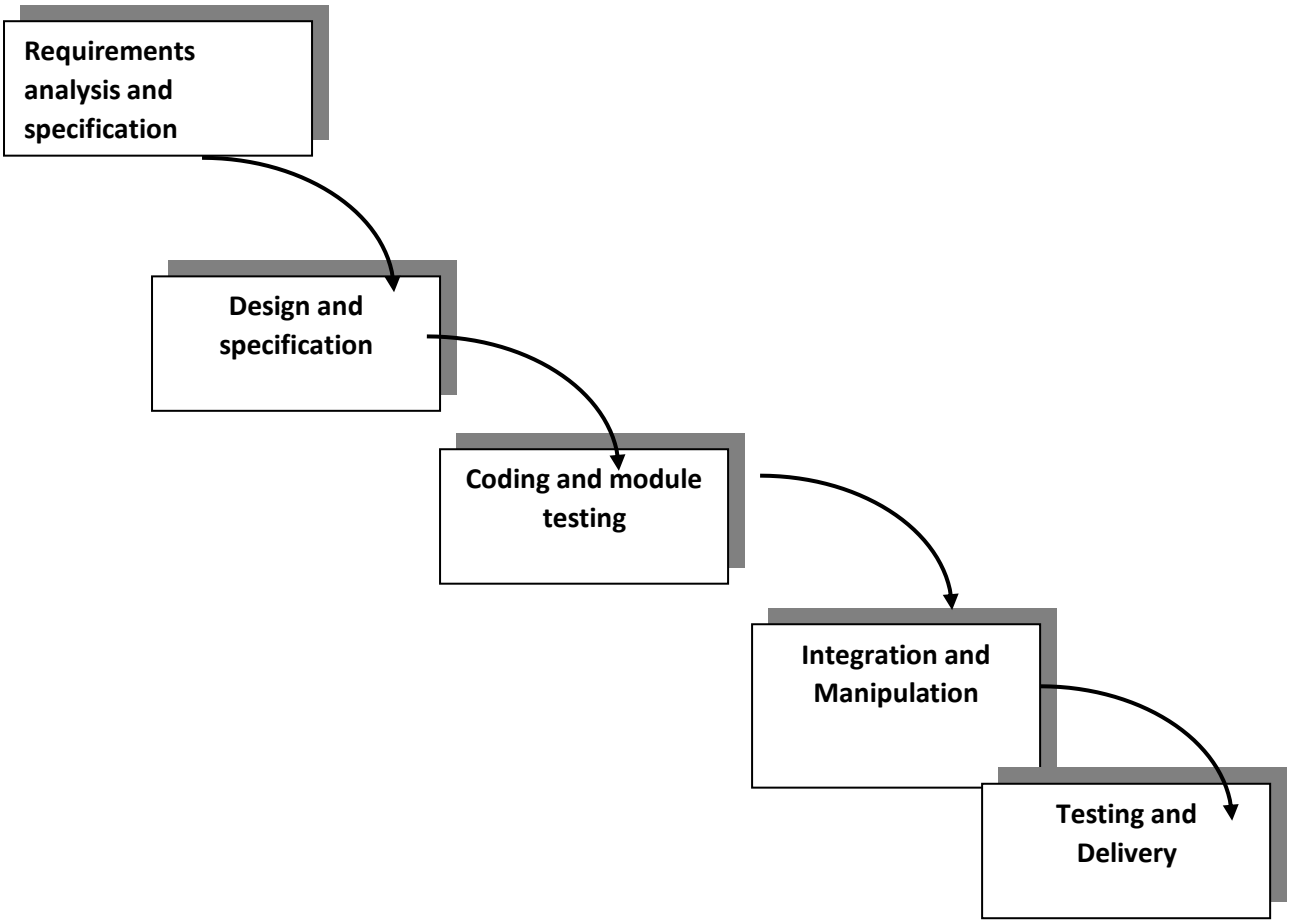
g). The Linear Sequential Model: The Linear Sequential Model, the Classic life cycle, or the Waterfall model suggests a systematic and sequential approach to the software development. The very first phase starts from the system level and progresses through analysis, design, coding, testing, and support. A pictorial view of opted model to develop the software is depicted in figure.

Software engineering paradigm is nothing but a life cycle model which is applied to give shape to the project. Without a life cycle model, a project is bound to fail because unless and until a planned vision is before hand, the situation is chaotic. There are numerous life cycle models which are available to software developers. One of the renowned and probably the oldest S.D.L.C model is Water Fall Model. This model as the name implies, is a phased model. Each phase has well defined starting and ending boundaries. One phase cannot begin unless it's previous phase is completed.

Waterfall Model:

In our case, we have decided to use waterfall model because the requirements have been well defined and stable which is the prime criteria for applicability of waterfall model. Also, the time allowed for the project is limited; there is no scope for applying an evolutionary model. Activities have to be distributed throughout the life cycle effectively. This model is easy to understand and easy to explain to the customer. With a small team size of two persons only, this model seems appropriate to us.

The following diagram depicts the various phases of the waterfall model: -



Software Requirement Specification (SRS)

1) Overview

The software requirement specification (SRS) is very important part of the software building process, which describes the actual user level requirement from technical point of view i.e., what the user exactly wants? Or For what purpose we are making every thing? The objective of preparing the software requirement specification is to represent the requirements of the software in such a manner that ultimately leads to successful software implementation. It is the result of the analysis process of the software development. It should contain all the data the software is going to process, the function it will provide, and the behavior it will exhibit. This Software Requirements Specifications (SRS) is defined in IEEE Std. 830-1993, IEEE Recommended Practice for Software Requirements Specifications. The document is organized in the following structure:

- Introduction
- Information Description
- Functional Description
- Behavior Description
- Validation Criteria
- Bibliography
- Appendix

a). Introduction: The introduction section describes the goals and objective the software under going development in context of computer based system. It mainly deals with the software scope i.e. it will bind the software application domain.

b). Information Description: This section of the SRS provides a detailed of the problem that the software must solve. It should describe the core of the software – i.e. The Data.The data or information the software is going to work on is the most basic part of the software. The description of each data or information entity is described here. It also gives details of the relationships between the data elements of the software. The information description helps the software designers in their designing purpose.

c). Functional Description: This section of the SRS describes the each functions required to solve the problem. It emphasizes on the core of the software on which the data will be processed – i.e. The Function. It also includes the process specification of each function, design constraints, and performance characteristics. The DFD or any other graphical diagram can also be added to describe the functionality of the system.

d). Behavioral Description: This section of the SRS describes the behavior of the software will exhibit. It is based on definition of the events and the operations that it will perform because of events.

e). Validation Criteria: This section of the SRS contains the details of the tests that should be performed to validate functions, performance, and behavior of the software. It is one of the most important aspect of the software which decides how much robust our software is.

f). Bibliography: This section contains references to all the related documents related with the software. This may include any technical document, standards document or software engineering paper.

g). Appendix: This section is supplementary and can include the matters that are important for the software development. It may include the statistical data, graphs or algorithm details.

Analysis Modeling

Data Modeling

Data modeling is very important part of the analysis it mainly deals with the data object and their relationship. With the help of data model one can easily identify the primary objects to be process by the system. Data model makes Entity-Relationship (ER) diagram to describe the relationship between entities. The data model considers the data independent of the processing that transform the data.

Data modeling answers a set of specific questions that are relevant to any data processing application. What are the primary data objects to be processed by the system? What is the composition of each data object and what attributes describe the object? What do the objects currently reside? What are the relationship between each object and other objects? What are the relationships between the objects and the processes that transform them?

a). Data Objects and Attributes

The data model consists of the data objects and attributes that describe the data object. A data object is the representation of any composite information that is processed by computer software. A data object can be an external entity, a thing, an occurrence or event, a role, a structure etc. A data object encapsulates the data only—there is no reference within a data object to operations that act on the data. So the data object can be a table.

Attributes define the properties of a data object and take on one of three different characteristics.

1. Name an instance of the data object
2. Describe the instance
3. Make reference to another instance in another table.

Also one or more attribute can be defined as identifier i.e. - **Key**. The key is used to make the relationship with the attribute of another data object. It may or may not be unique. In our context, we can say that as our project is RDBMS, so here relationship between various data object is essential to get required information and result the user want. So key has very important role here to make references as well as referential integrity and to prevent from data redundancy.

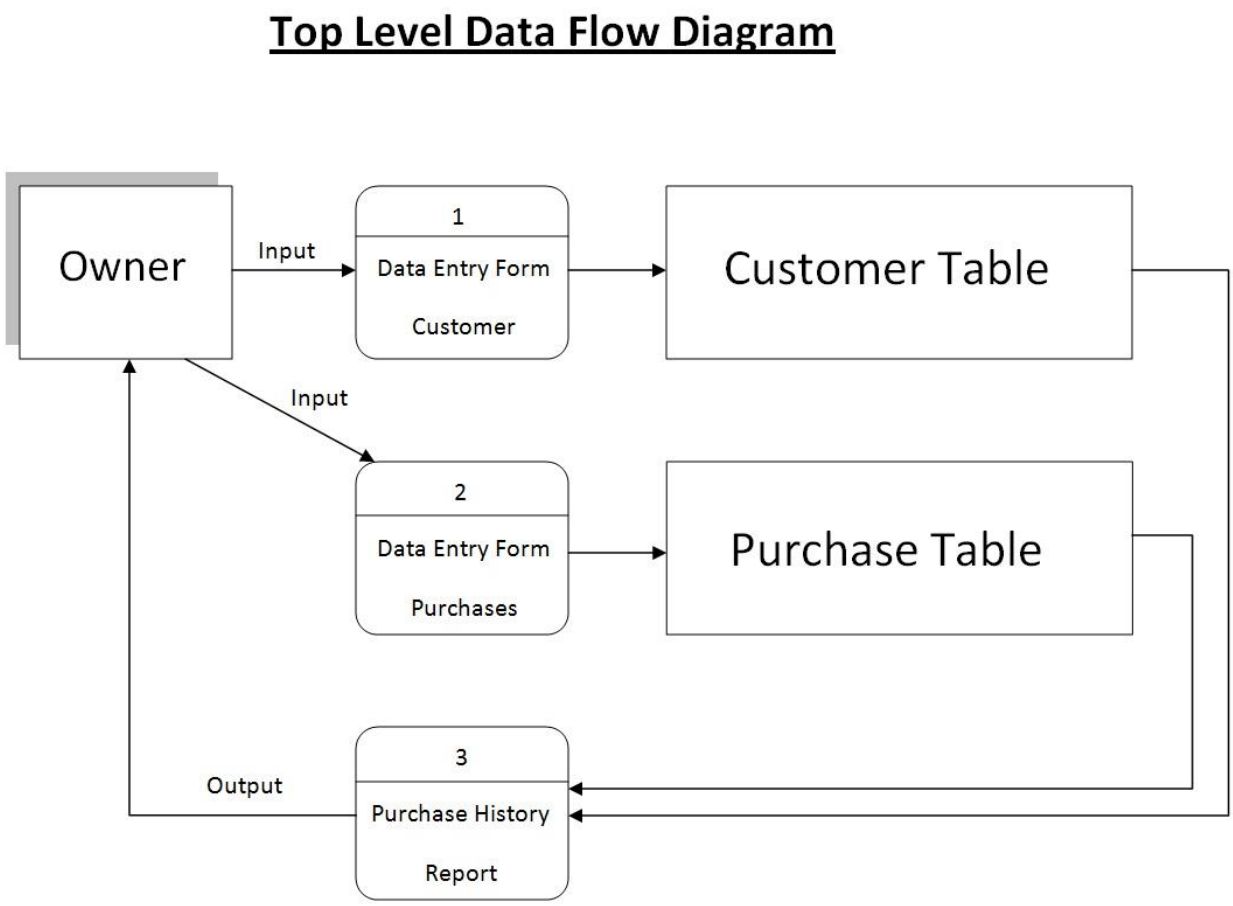
b). Cardinality and Modality

Cardinality: Cardinalitydefines the maximum numberof objects that can be participating in a relationship. According to Tillman, the cardinality is the specification of the number of occurrences of one object that can be related to the

number of occurrences of another object. Taking into consideration all combinations of 'one' and 'many' two objects can be related as:

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-One (M:1)
- Many-to-Many (M:N)

Data Flow Diagram



Flow Chart

Requirement Gathering

Requirement gathering refers to the collection of information pertinent to system project. To get information, we read books or reports; we go through company, collected forms from the company for later analysis and interview people. Interviewing is an important skill and we apply to the company manager, the teller and even the customers.

As information is collected we documented the important aspects so it can be referred to later on for this purpose we used charts and tables.

The principle methods of obtaining facts include:

1. Interviewing personnel.
2. Observing activities.
3. Use of questionnaires or by inspections.

In general, the stages of investments are as follows:

1. Define the problem.
2. Plan the project.
3. Record the facts.
4. Verify the records.

Creation of Alternatives

Ones we get the clear idea of the problem, we begin to create some solutions usually begin to form while the initial research is going on. Then after completing the research, we choose the most promising alternatives and develop them.

In order to create alternatives, we have a broad background and familiar with the many different types of equipment that can be applied to the problem and familiar with the various types of procedures that can be used. From this background, we can develop an alternative similar to one that other company is using or can create special or unique solutions to the company problem.

It is important to realize that the solutions at this stage are not developed in detail. The procedures develop here are not specific. Although a general logic flow is created for each alternative, specific steps will be determined during the system design phase.

After creation of alternatives completed, now we are able to proceed with the Context Analysis Diagram and further Data Flow Diagram for our system.

Requirement Gathering

Requirement gathering refers to the collection of information pertinent to system project. To get information, we read books or reports; we go through company, collected forms from the company for later analysis and interview people. Interviewing is an important skill and we apply to the company manager, the teller and even the customers.

As information is collected we documented the important aspects so it can be referred to later on for this purpose we used charts and tables.

The principle methods of obtaining facts include:

4. Interviewing personnel.
5. Observing activities.
6. Use of questionnaires or by inspections.

In general, the stages of investments are as follows:

5. Define the problem.
6. Plan the project.
7. Record the facts.
8. Verify the records.

A part from collecting facts as indicated above, there are other specific points to be considered in a system analysis, as follows:

a) Collect specimen of all documents used in the system.
b) From the documents collected prepare.
    i. Document analysis from input.
    ii. Document analysis from output.
    iii. File analysis from.
    iv. Output analysis chart.

Creation of Alternatives

Ones we get the clear idea of the problem, we begin to create some solutions usually begin to form while the initial research is going on. Then after completing the research, we choose the most promising alternatives and develop them.

In order to create alternatives, we have a broad background and familiar with the many different types of equipment that can be applied to the problem and familiar with the various types of procedures that can be used. From this background, we can develop an alternative similar to one that other company is using or can create special or unique solutions to the company problem.

It is important to realize that the solutions at this stage are not developed in detail. The procedures develop here are not specific. Although a general logic flow is created for each alternative, specific steps will be determined during the system design phase.

After creation of alternatives completed, now we are able to proceed with the Context Analysis Diagram and further Data Flow Diagram for our system.

System Design

The system is designed based on the objectives set for the new information system in the strategic plan. Our outermost objectives is as follows:

1) Increasing the company profit
2) Securing the customer intact with the company
3) Improving customer services
4) Understanding user requirement
5) Suggesting a logical organization of data
6) Monitoring and controlling decision
7) Securing a larger market share.

Up to the time of system proposal, we have concerned them with the local design of the system. Although we have decided that specific software should be purchased as well as interface like C++. After the system proposal is agreed to, we concern them with physical design of the system. Here we create, detailed specification for every aspect of the system. This includes form design, file organization, procedural steps and so on.

The following are the major steps in system design:

1) Specification of data elements, records and files
2) Specification in input forms, data preparation formats and identification of personnel who will complete them.
3) Specification of system output
4) Development of system flow charts
5) Development of feedback and control procedures
6) Development of program specifications
7) Development of operating specifications
8) Resource planning and implementation:
    - Phasing out the existing system
    - Training user or personnel
    - System review and parallel operation

Training of Personnel Involved with the System

Those who are directly or indirectly related with system development work must know in detail what their roles will be, how they can make the efficient use of the system and what the system will do or not do for them. Here both the employees and users need training.

System Operator's Training:

Running of the system successfully depend on the personnel working on that system. So their training must ensure that they are able to handle all possible operations, both routine and extra-ordinary in nature.

Keeping the above objectives in mind system operator's are trained with some fundamental topics as:

- How to turn the equipment on and use it,
- How to power off and a knowledge what constitutes normal operation,
- Different types of malfunctioning, how to recognize them, and what steps should be taken whenever they arise.
- As part of their training, operators are given both a troubleshooting list that identifies possible problems and remedies for them, as well as the names and telephone numbers of individuals to contact when unexpected or unusual problems arise.
- Familiarization with run procedures, which involves working through the sequence of activities needed to use a new system on an ongoing basis.

User's Training:

User training deals with the operations of the system itself, with proper attention given to the data handling techniques.

Users are trained in methods of entering transactions, editing, formulating inquiries, and view reports. Users are familiarized with simple systems maintenance activities.

Conversion Methods

Parallel System: For conversion of old system to new one we employ parallel system to run both the system in parallel. Under this approach, users continue to operate their old system as well as they also start using the new system.

This method is safest one because it ensures that in case of any problems in using the new system, the organization can still fall back to the old system without loss of time and money.

After the system conversion is completed using parallel system, the conversion plan starts. During the pre-implementation stages, when the conversion is being planned, we were prepared a list of all tasks, including the following:

- List of all files that are required during conversion
- Identify all data required to build new files during conversion
- Identify all controls to be made during conversion
- Assign the responsibility for each task.

And now at this stage we convert all the above conversion plans into operation one by one.

Training Methods

In-house Training: The main advantage of offering in-house training is that instructions can be tailored according to the requirements of the organization.

Organization can involve more personnel in the training program because this type of training method is more economical.

System Maintenance

System maintenance is actually the implementation of the post-implementation review plan. So, at this stage we implement the following maintenance area:

i) Administrative Maintenance: The following activities are maintained under this maintenance:

- User objective maintenance: This is an extremely critical area since it may be possible that over the period of time either the system does not meet the initial objectives of the user or the user objectives get changed as a result of the changes in the overall objectives of the organization.
- Operating costs and benefits maintenance: Under this maintenance, current budgets are designed to manipulate the costs and savings of the system.

ii) Personnel Requirement Maintenance: Under this maintenance, all activities involving system personnel and staff members associated with the system are maintained. That is,

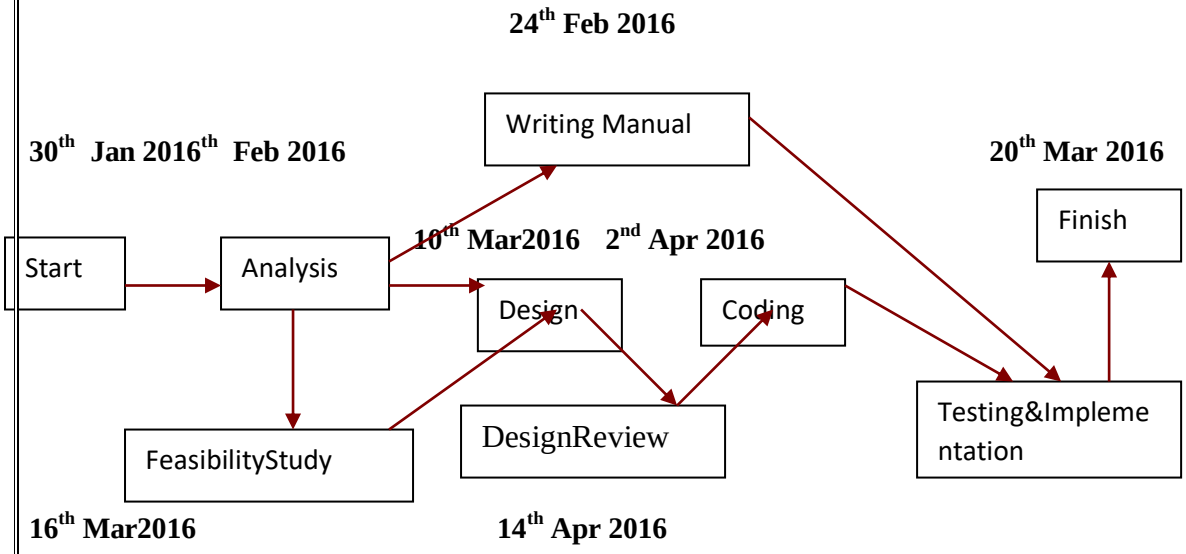
- Maintained the performance of personnel objectives with current performance levels.
- Maintained training performance through testing and conducting interviews.

iii) **Hardware Maintenance:** The hardware of the new system is also maintained including servers, terminals, and communication networks. The main target is a comparison of current performance specifications with design specifications.

For implementing the overall post-implementation plan, actually we employ little efforts, because most of essential tasks that are maintained in this stage are already performed with more efficiently. For example:

- During system development the user's requirements was defined more accurately.
- System documentation was prepared in a better way.
- Processing logic was also designed using more effective way.
- Existing tools and techniques are better utilized.
- System engineering processes are managed effectively.

Pert Chart


24th Feb 2016
Writing Manual
30th Jan 2016th Feb 2016
20th Mar 2016
Finish
10th Mar2016 2nd Apr 2016
Start
Analysis
Design
Coding
Testing&Impleme
ntation
DesignReview
FeasibilityStudy
16th Mar2016
14th Apr 2016

Feasibility Study

Introduction

A Feasibility Study determines whether a project is worth doing. The process followed for making this determination is called a Feasibility Study. This type of study determines whether a project can and should proceed. Once it has been determined that a project is feasible, the analyst can proceed and prepare the project specifications that finalize the project specification.

The following are the various types of feasibility studies that can be undertaken.

Technical Feasibility

This is concerned with specifying the equipments and the software to satisfy the user requirements. The technical needs of the system vary considerably but might include:

- The facility to produce outputs in a given time.
- Response time under certain conditions.
- Ability to process a certain volume of transactions at a specified speed.
- Facility to communicate data to a distant location.

Technical feasibility centers on the existing computer system, hardware, software etcetera and to what extent it can support the system. In examining the technical feasibility, the configuration of the system is given more importance than the actual hardware. The configuration should provide the complete picture of the system requirements, for example how many workstations are required and how these units are interconnected so that they would operate smoothly, etcetera. The result of the Technical Feasibility Study is the basis for the documents against which dealer and manufacturer can make bids. Specific hardware and software products can then be evaluated keeping in view the logical needs.

Economic Feasibility

Economic analysis is the most frequently used method for evaluating the effectiveness of a new system. More commonly known as cost/benefit analysis, the procedure is to determine the benefits and savings that are expected from a candidate system and compare them with costs. If benefits outweigh costs, then the decision is made to design and implement the system.

It is not done to analyze the new system. Using a Gantt Chart schedule and part chart. We assumed that the benefit of the project is greater than the cost. So if we can develop the project easily then it is used for the evaluation of the proposed. We calculate the cost/benefit analysis and we assume that the benefit is feasible so we start developing the project. It is an analysis of the cost to be incurred in the system and benefits the derivable from the system. An economic Feasibility Study should demonstrate the net benefit of the proposed course of action in the context of direct and indirect benefits and costs to the organization and to the public as a whole. It should be required for both pilot and long-term activities, plans and projects.

Operational Feasibility

It determines how acceptable the software is within the organization. The evaluations must then determine the general attitude and skills. Such restriction of the job will be acceptable. To the users are enough to run the proposed budget, hence the system is supposed to the feasible regarding all except of feasibility. In operational feasibility, we attempt to enusre that every user can access the system easily. We develop a menu that users can easily access and we provide shortcut keys.

We show a proper error message when any mistakes are made in the program. We provide help and a guideline menu to help the user.
Changes in the ways individuals are organized into groups may then be necessary and the groups may now compete for economic resources with the needs of stabilized ones by converting a number in a file in software.

Behavioral Feasibility

Normal human psychology of human beings indicate that people are resistant to change and computers are known to facilitate change. Any project formulations should consider this factor also. Before the development of the Project titled "Delhi Metro", the need to study the feasibility of the successful execution of the project was felt and thus the following factors are considered for a Feasibility Study.

1. Need Analysis.
2. Provide the users information pertaining to the preceding requirement.

Feasibility Study Report

The result of the Feasibility Study provides us with the following facts:

- The automated system would increase the efficiency of the system.
- The automated system would increase customer's satisfaction.

- The automated system has many requirements such as Efficiency cost effectiveness, prompt service, Reliability.
- The automated system would add to the security features of the system
- The automated system should be simple to use, incorporate all necessary services and maintainable.
- This will cause some changes in the organization. These are:
-

Change in staffing policies: present employees will need to be sent for training.

An outline of a feasibility study is as follows:

1) Statement about the objective of the problems
2) Terms of reference
3) Analysis of existing system
    - Data gathering
    - Data presentations
    - List of files and records required
    - Preparation of flow charts
    - Cost estimates
4) Analysis of alternatives meeting similar requirements. For each alternatives proposed:
    - Overview
    - Decision
    - Information flows
    - Technical details
    - Operational details

- Impact on the organization
- Total cost and benefits

5) Determine the main output requirements
6) Effects on company operations
7) Financial effects
8) Achievement of company objects
9) Conclusion.

Testing

Testing is the filter to catch defects before they are "discovered" by the customer or end user. Every time a customer runs the program, he/she generates a "test-case". We tried the test cases to find the defects first, since software development is a human activity and there may be potential errors. So, testing before the product is delivered helped us to assure the quality and saved resources.

a). Testing Objective

Objective of testing is to find errors, demonstrate software functions that satisfy the specification reduce the number of errors detected by customer, and have "confidence" in the software system. A good test always tries to discover undetected errors and is successful when errors are found since zero defects is not possible. There is always a condition or usage that can lead to an incorrect behavior.

b). Testing Steps

We started testing each individual new component and worked out as unit test, integration test, high order test, customer acceptance testing and different testing techniques are used at different times in the process.

c). Testing Techniques

Following testing techniques are used:-

i). White Box Testing

In white box testing we exercises the internal logic of a program, examine internal program structure and derived tests from an examination of the program's logic, tested internal paths and working of the software. To perform this we used to develop test cases for unit and integration testing.

ii). Black Box Testing

In black box testing we exercises the input and output requirements of a program, tested by conducting specifications of requirement for which the software should do. This test is derived from the I/O specification and used in most functional tests.

iv). White Box Strategies

White box testing strategies uses the control structure of the program and design to derive test cases.

Basis Path Testing

Test cases are derived to exercise the basis set that guarantee that every statement in the program is executed at least once during testing. It is applied in series of steps.

Test case for login button of the login form:

Test case 1: All the following combinations are tested

Password is valid and Username is valid

Password is valid and Username is invalid

Password is invalid and Username is invalid

Password is invalid and Username is valid

Expected results:

No error crossing this stage when Password & Username are both permissible combination of characters. In all other case error message is displayed.

Test case 2:Following input combination is tried where permissible set of characters for username & password are entered which are non-existent in table.

Password is valid, existent and Username is valid, existent

Password is valid, existent and Username is valid, nonexistent

Password is valid, nonexistent and Username is valid, existent

Password is valid, nonexistent and Username is valid, nonexistent

Expected Results:

If no entry into the system then error message displayed. And user is linked to the registration page.

Test Case 3:To test this both username & password to be entered must be valid & existent in the registration table.

Password = Valid, Existent ; and

Username = Valid, Existent

Similarly all the other internal paths & logic can be tested.

v). Control Structure Testing

Condition testing

Condition testing is a test case design method that exercises the logical conditions contained in a program module. If a condition is incorrect, then at least one of the conditions is incorrect. So, types of errors in a condition include Boolean operator error (incorrect/missing/extra Boolean operators), Boolean variable error, Boolean parenthesis error, relational operator error and arithmetic expression error. It mainly focuses on testing each condition in the program. The condition testing strategies to be employed are:

Branch testing:

For a compound condition C, the true and false branches of C and every simple condition in C need to be executed at least once.

Domain testing:

It requires three or four tests to be derived for a relational expression for a relational expression of the form

E1 <relational operator> E2

Three tests are required to make the value of E1 greater than, equal to, or less than that of E2.

Boolean Expression testing:

A condition without relational expression with n variables, all of 2^n possible tests are required provided n > 0. This strategy can detect Boolean operator, variable, and parenthesis errors, but it is practical only if n is small.

Loop testing

Loops are cornerstone for the vast majority of all algorithms implemented in software. Loop testing focuses exclusively on the validity of loop constructs. Testing approach for different types of loops is listed below.

Simple loops:

The following set of tests will be applied to simple loops, where n is the maximum number of allowable passes through the loop.

- Skip the loop entirely
- Only one pass through the loop
- Two passes through the loop
- m passes through the loop where m < n
- n – 1, n, n + 1 passes through the loop.

Nested loops:

Simple loop testing strategy if applied to the nested loops will increase the no. of possible tests geometrically. Approaches for nested loops are:

- Start at the innermost loop. Set all other loops to minimum values
- Conduct simple loop tests for the innermost loop while holding the outer loops at their minimum iteration parameter values. Add other tests for out-of-range or excluded values.
- Work outward, conducting tests for the next loop, but keep all other outer loops at minimum values and other nested loops to typical values.
- Continue till all loops have been tested.

Concatenated loops:

When concatenated loops are independent then approach for simple loop was applied & if they happen to be dependent the nested loop approach was applied.

Unstructured loops:

This class of loops was redesigned to reflect the use of the structured programming constructs.

vi). Black Box Strategies

Black Box testing strategy focuses on the functional requirements of the software without any regard to the internal structure. It is used in most system level testing where functional, performance, recovery and security & stress are main issues. This applied strategy finds errors in incorrect or missing functions (compared to white box), interface errors, errors in external data structures and in behavior performance problems (Combinations of input make it behave poorly). It also finds errors in initialization and termination (Sensitive to certain inputs (e.g. performance). It is performed much later in process than white box...

A test strategy that uses every possible input condition as a test case would be ideal but is not possible. We apply following Black-Box Testing Strategies.

Graph Based Testing

The first step in black box testing is to understand the objects that are modeled in software and the relationships that connect the objects. Once this has been accomplished, the next step is to define a series of tests that verify all objects that have the expected relationship with another.

Equivalence Partitioning It is a black-box testing method that divides the input domain of a program into classes of data from which test cases can be derived. It reduces, by more than one, the number of test cases that must be developed since one test case might uncover a class of errors. It covers a large set of other possible test cases hence helps reduce the number of inputs. Incorrect processing of all integer number is detected.

Identifying Equivalence Classes:Each input condition (a sentence or phrase in the specification) is taken & divided into 2 or more classes as valid equivalence class and invalid equivalence class.

Rules for Equivalence Classes:

Range - If an input condition specifies a range (i.e. n is an integer from 1 to 10) then

i) 1 valid (if 1<= n and <= 1000)
ii) 2 invalid (if n < 1 and > 1000)

Specified Value - An input condition specifies a specific value e.g. 10 character string then

i) 1 valid (10 character string)
ii) 2 invalid (any character string less than 10)

Value Set - If the input specifies a set of valid values, then

i) 1 valid condition within the set.
ii) 1 invalid condition outside the set.

Boolean -If an input condition specifies a "must be" situation (e.g. either Yes or No) then

i) 1 valid (if yes).
ii) 1 invalid (else No)

Boundary Value Analysi

It is a test case design technique that complements equivalence partitioning, by selecting test cases at the "edges" of the class. Greater no. of errors tends to occur at the boundaries of the input domain rather than in the "Center". Rather than any element in class, BVA selects tests at the edge of the class. In addition to input condition, test cases can be derived for output conditions. Similar to Equivalence partitioning, BVA, equivalence classes are identified first then boundaries.

Some guidelines for BVA:

1. If an input condition specifies a range bounded by values a & b then test cases will be designed with values a and b and just above and just below a & b.

2. If an input condition specifies a number of values, test cases will be developed that exercise the minimum and maximum numbers. Values just above and just below minimum and maximum are also tested.

These 2 guidelines are also applied to the output conditions. Designing test case to exercise the data structure at its boundary will test boundaries prescribed on internal program data structures.

Testing Strategies for Web Application

The approach of web app testing adopts the basic principle for all Software testing and applies a strategy and tactics that have been recommended for object oriented system .the following steps

Summarizes the approach:

1). The content model for the web app is reviewed to uncover errors.

This is like copyediting

2). The design model for the web application as reviewed to uncover navigation error.

Use cases derived as part of the analyst activity, allow a web engineer to exercise each usage scenario against the architectural and navigational design. These non executable test help uncover error in navigation.

3). Selected processing component and web page is unit tested.

When web apps are considered, the concepts of the unit changes, each web page encapsulated in itself content navigation link as Well As script, form and applet (processing element). It is not always possible or practical to test each of these characteristics individually.

4).The architecture is constructed and integration tests are Conducted.

The strategy for integration testing depend upon the architecture that has been chosen for the web application

5). The assembled web application is tested for overall functionality and content delivery.

Like conventional validation, the validation of web based systems and application focuses on user visible action and user recognizable output from the system. To assist in the derivation of validation tests the tester should draw upon use cases the use cases provides a scenario that has high likelihood of uncovering errors in user interaction requirement.

6). Theweb application is implemented in a variety of different Environmental configuration and is tested for compatibility with each configuration.

7).The web application is tested by controlled and monitoredPopulation of end user.

Finally with the modular concept inside the application it is being also tested for its Reliability: The system reliability will be insured through data integrity rules built into the database at the backend and the system rules built into the front-end application. The system will take assurance from the user before making any changes permanent.

d). Maintainability

The system has been designed taking care of modularity. Faults in the system can be traced to modules and thus

e). Validation Checks

This will be as such to maintain consistent and persistent Information on the web when most of the time the project has to Deal With uploads so a minor error will down the impression of the Company. Therefore, validation checks by software itself using the Flavors of JavaScript and manual checks are also necessary as: -

1. Correct entry of data in the form.

2. Correct updating of question and with the most suitable answer etc.

Maintenance

Software maintenance is a set of software engineering activities that occur after software has been delivered to the customer and put into operation. A primary goal of software engineering is to improve the ease with which changes can be made. The software maintenance can be defined by four activities:

- Corrective Maintenance
- Adaptive Maintenance
- Perceptive Maintenance or Enhancement
- Preventive Maintenance or Reengineering

The software documentation is very important job regarding maintenance, because it is a big problem for the user to make any change in working version of software without consent of the developer.

In out context as we developed and designed the core software for a website having the maximum possibility of change we have kept the copy of documentation on the site .As well as are always ready to edit it when any major change is being done at the design and coding part.

Screen

Customer Information

```
                    AFFWA CANTEEN MANAGEMENT

CONSUMER INFORMATION


 NAME:Poonam Rajput

 RANK:2

 SERVICE NO.:123

 SMART CARD NO.:12345

 PHONE NO.:9412511423

 ENTER BILL NO.:3
```

Main Menu

```
1.MENU & PURCHASE

2.DETAILS

3.NEW CUSTOMER

4.QUANTITY AVAILABLE

5.EXIT

ENTER YOUR CHOICE:1_
```

Product List

```
BISCUTE                                                    10
PEPSODENT                                                  36
DENIM POWDER                                               110
MUNCH                                                      8
PARK AVENUE PERFUME                                        110
COCONUT HAIR OIL                                           90
TITAN WATCH                                                1199
LUX SOAP                                                   17
TIDE                                                       107
REEBOK SHOES                                               1499

DO YOU WANT TO PURCHASE(Y/N):
_
```

Invoice Bill

```
BISCUTE                                  2                        Rs.20
PEPSODENT                                2                        Rs.72
DENIM POWDER                             1                        Rs.110
MUNCH                                    5                        Rs.40
PARK AVENUE PERFUME                      1                        Rs.110
COCONUT HAIR OIL                         2                        Rs.180
TITAN WATCH                              2                        Rs.2398
LUX SOAP                                 5                        Rs.85
TIDE                                     1                        Rs.107
REEBOK SHOES                             3                        Rs.4497
================================================================================
TOTAL:                                   24                       Rs.7619
THANK YOU .                         VISIT US AGAIN
```

Canteen Details

```
                    CANTEEN DETAILS

#TIMING:

 11 AM TO 02 PM
 03 PM TO 05 PM

#CANTEEN WILL CLOSED ON TUESDAY

#GOODS ONCE SOLD WILL NOT BE EXCHANGED/REPLACED

#TAKE YOUR SMART CARD WITH YOU

#COLLECT YOUR SMART CARD & BILL AFTER PAYMENT

#DO NOT BREAK ANY CANTEEN ITEMS
```

Stock Details

```
DENIM POWDER                                109

MUNCH                                       995

PARK AVENUE PERFUME                         499

COCONUT HAIR OIL                            398

TITAN WATCH                                  98

LUX SOAP                                    995

TIDE                                        999

REEBOK SHOES                                297
*******************************************************************************
```

Coding

```
/*              &&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&
                & COMPUTER SCIENCE PROJECT WORK&
                &&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&

            SOFTWARE FOR THE USE OF CANTEEN DEPARTMENT

////INTRODUCTION////
//////////////////////////
//NAME:- GOURAV.K.CHOUDHARY /
//CLASS:- XI 'D'          /
//ROLL NO.:- 33           /
//////////////////////////
####################
#CANTEEN MANAGEMENT#
####################       */

/*## HEADER FILES ##*/
#include<iostream.h>
#include<conio.h>
#include<string.h>
#include<stdio.h>
#include<process.h>
#include<iomanip.h>
#include<dos.h>

main()
 {
  clrscr();
unsignedint sum,add,d,i,j,e,f,g,h,k,l,w,x,y,z;
unsignedint bir=10,per=36,der=110,mur=8,par=110,cor=90,twr=1199, lur=17,tir=107,
              rer=1499;
int
qubis=1000,abis=0,qupep=1000,apep=0,quden=110,aden=0,qumun=1000,amun=0,qu
```

```
perk=500,aperk=0,qucoc=400,acoc=0,qutit=100,atit=0,qulux=1000,alux=0,qutid=100
0,atid=0,quree=300,aree=0;
unsignedint m=0,n=0,o=0,p=0,q=0,r=0,s=0,t=0,u=0,v=0;
char name[50],rank[10],pu,c[20],a[15],b[15],ok;
  clrscr();
 printf("\n
++++++++++++++++++++++++++++++++++++++++*++++++++++++++++++++
+++++++++++++++++");
 printf("\n +                          ***                        +");
 printf("\n +                         *****                        +");
 printf("\n +                        *******                        +");
 printf("\n +                       *********                        +");
 printf("\n +                      ***********                        +");
 printf("\n +                     *************                        +");
 printf("\n +                    ***************                        +");
 printf("\n +                   *****************                        +");
 printf("\n +                  *******************                        +");
 printf("\n +                 *********************                        +");
 printf("\n +                ************************                        +");
 printf("\n +               **************************                        +");
 printf("\n +              ****************************                        +");
 printf("\n +             ******************************                        +");
 printf("\n +            ********************************                        +");
 printf("\n +           **********************************                        +");
 printf("\n +          ************************************                        +");
 printf("\n +         ***************************************
+");
 printf("\n +        *****************************************
+");
 printf("\n +       ******************************************
+");
 printf("\n +      *******************************************
+");
 printf("\n +     ********************************************
+");
 printf("\n +    **********************************************
+");
 printf("\n +
*********************WELCOME************************        +");
 printf("\n +    **********************************************
+");
```

```
printf("\n +          ********************************************
+");
printf("\n +          *******************************************
+");
printf("\n +          ******************************************
+");
printf("\n +            ****************************************
+");
printf("\n +             ***************************************          +");
printf("\n +              **************************************         +");
printf("\n +               *************************************        +");
printf("\n +                ************************************       +");
printf("\n +                 ***********************************      +");
printf("\n +                  **********************************     +");
printf("\n +                   *********************************    +");
printf("\n +                    ********************************   +");
printf("\n +                     *******************************  +");
printf("\n +                      ******************************  +");
printf("\n +                       *****************************  +");
printf("\n +                        ****************************  +");
printf("\n +                         ***************************  +");
printf("\n +                          **************************  +");
printf("\n +                           *************************  +");
printf("\n +                            ************************  +");
printf("\n +                             ***********************  +");
printf("\n +                              **********************  +");
printf("\n +                               *********************  +");
printf("\n +                                ********************  +");
printf("\n +                                 *** BY:- iCBSE.com  +");
printf("\n
++++++++++++++++++++++++++++++++++++++++++*++++++++++++++++++++++++++++++++++");
 getch();
 clrscr();
 cout<<"\n\n\t    w        w   e e e e e e e   l       ";
 cout<<"\n\n\t    w    w    w   e           l       ";
 cout<<"\n\n\t    w   w w   w   e           l       ";
 cout<<"\n\n\t    w   w  w   w   e e e e e e e   l       ";
 cout<<"\n\n\t    w   w   w   w   e e e e e e e   l       ";
 cout<<"\n\n\t    w w    w w   e           l       ";
 cout<<"\n\n\t    w w     w w   e           l       ";
 cout<<"\n\n\t    w        w   e e e e e e e   l l l l l l l";
 cout<<"\n\n\n\n  c c c c c c c    o o o o o o o    m         m   e e e e e e e  ";
 cout<<"\n\n  c          o        o   m m      m m   e ";
 cout<<"\n\n  c          o        o   m  m     m  m   e   ";
 cout<<"\n\n  c          o        o   m   m    m   m   e e e e e e e    ";
```

```
 cout<<"\n\n  c              o          o    m    m    m    m    e e e e e e e e   ";
 cout<<"\n\n  c              o          o    m    m  m    m    e     ";
 cout<<"\n\n  c              o          o    m    m m    m    e    ";
 cout<<"\n\n   c c c c c c c    o o o o o o o    m     m     m    e e e e e e e e   ";
 getch();
 clrscr();
 main:;
 cout<<"\n\n\n\t\t\t     AFFWA CANTEEN MANAGEMENT";
    cout<<"\n\n\nCONSUMER INFORMATION";
    cout<<"\n\n\n\n NAME:";
    gets(name);
    cout<<"\n RANK:";
    gets(rank);
    cout<<"\n SERVICE NO.:";
    cin>>a;
    cout<<"\n SMART CARD NO.:";
    cin>>b;
    cout<<"\n PHONE NO.:";
    cin>>c;
    cout<<"\n ENTER BILL NO.:";
    cin>>i;
    again:;
    clrscr();
    cout<<"\n\n\n\n\1.MENU & PURCHASE";
    cout<<"\n\n\n\2.DETAILS";
    cout<<"\n\n\n\3.NEW CUSTOMER";
    cout<<"\n\n\n\4.QUANTITY AVAILABLE";
    cout<<"\n\n\n\5.EXIT";
    cout<<"\n\n\nENTER YOUR CHOICE:";
    cin>>d;
    clrscr();
switch(d)
 {
case 1:
    cout<<"\nITEMS\t\t\t\t\t\t\t      RATE(RS)";
    cout<<"\n\n\nBISCUTE\t\t\t\t\t\t\t\t10  ";
    cout<<"\n\nPEPSODENT\t\t\t\t\t\t\t36";
    cout<<"\n\nDENIM POWDER\t\t\t\t\t\t\t110   ";
    cout<<"\n\nMUNCH\t\t\t\t\t\t\t\t8  ";
    cout<<"\n\nPARK AVENUE PERFUME\t\t\t\t\t\t110   ";
    cout<<"\n\nCOCONUT HAIR OIL\t\t\t\t\t\t90  ";
    cout<<"\n\nTITAN WATCH\t\t\t\t\t\t\t1199   ";
    cout<<"\n\nLUX SOAP\t\t\t\t\t\t\t17   ";
```

```
cout<<"\n\nTIDE\t\t\t\t\t\t\t\t107   ";
cout<<"\n\nREEBOK SHOES\t\t\t\t\t\t\t1499   ";
start:
cout<<"\n\n\nDO YOU WANT TO PURCHASE(Y/N):\n";
cin>>pu;
   if(pu=='Y'||pu=='y')
   {switch(pu)
   case 1:
     clrscr();
     cout<<"\n\n\n\t\t\t\tPURCHASE LIST";
     bis:;
     cout<<"\nBISCUTE(MAX 30): ";
     cin>>m;

   if(m>30)
     {
      cout<<"Enter Quantity Smaller than 30";
     getch();
   goto bis;
     }
   else
     abis=qubis-m;
     pep:;
     cout<<"\nPEPSODENT(MAX 2): ";
     cin>>n;
   if(n>2)
     {
      cout<<"Enter Quantity Smaller than 2";
     getch();
   goto pep;
     }
   else
     apep=qupep-n;
     den:;
     cout<<"\nDENIM POWDER(MAX 2): ";
     cin>>o;
   if(o>2)
     {
      cout<<"Enter Quantity Smaller then 2";
     getch();
   goto den;
     }
   else
```

```
  aden=quden-o;
  mun:;
  cout<<"\nMUNCH(MAX 50): ";
  cin>>p;

if(p>50)
  {
   cout<<"Enter Quantity Smaller then 50";
  getch();
goto mun;
  }
else
  amun=qumun-p;
  park:;
  cout<<"\nPARK AVENUE PERFUME(MAX 2): ";
  cin>>q;

if(q>2)
  {
   cout<<"Enter Quantity Smaller then 2";
  getch();
goto park;
  }
else
  aperk=quperk-q;
  coco:;
  cout<<"\nCOCONUT HAIR OIL(MAX 2): ";
  cin>>r;

if(r>2)
  {
   cout<<"Enter Quantity Smaller then 2";
  getch();
goto coco;
  }
else
  acoc=qucoc-r;
  titan:;
  cout<<"\nTITAN WATCH(MAX 2): ";
  cin>>s;

if(s>2)
  {
```

```
    cout<<"Enter Quantity Smaller then 2";
   getch();

goto titan;
   }
else
   atit=qutit-s;
   lux:;
   cout<<"\nLUX SOAP(MAX 10): ";
   cin>>t;

if(t>10)
   {
    cout<<"Enter Quantity Smaller then 10";
   getch();
goto lux;
   }
else
   alux=qulux-t;
   tid:;
   cout<<"\nTIDE(MAX 2): ";
   cin>>u;

if(u>2)
   {
    cout<<"Enter Quantity Smaller then 2";
   getch();
goto tid;
   }
else
   atid=qutid-u;
   ree:;
   cout<<"\nREEBOK SHOES(MAX 4): ";
   cin>>v;

if(v>4)
   {
    cout<<"Enter Quantity Smaller then 4";
   getch();
goto ree;
   }
else
   aree=quree-v;
```

```
        cout<<"\n";
        clrscr();
        cout<<"\n\n\n\n\n\n\t\t\t\tAFFWA CANTEEN\n";
        cout<<"\n\n\t\t\t\t  CASH MEMO\n";
        cout<<"\n\nBILL NO.:"<<i;
        cout<<"\n\nNAME:";
        cout<<name;
        cout<<"\n\nRANK:"<<rank;
        cout<<"\n\nSERVICE NO.:"<<a;
        cout<<"\n\nSMART CARD NO.:"<<b;
        cout<<"\n\nPHONE NO.:"<<c;
        cout<<"\n\n\n\n\n";

cout<<"==========================================================
========================";
        cout<<"\nITEMS"<<setw(40)<<"QUANTITY"<<setw(35)<<"PRICE(Rs.)";

cout<<"==========================================================
========================";
      if(m>0)
         cout<<"\n\nBISCUTE"<<setw(38)<<m<<setw(28)<<"Rs."<<m*bir;
      if(n>0)
         cout<<"\n\nPEPSODENT"<<setw(36)<<n<<setw(28)<<"Rs."<<n*per;
      if(o>0)
         cout<<"\n\nDENIM POWDER"<<setw(33)<<o<<setw(28)<<"Rs."<<o*der;
      if(p>0)
         cout<<"\n\nMUNCH"<<setw(40)<<p<<setw(28)<<"Rs."<<p*mur;
      if(q>0)
         cout<<"\n\nPARK AVENUE
PERFUME"<<setw(26)<<q<<setw(28)<<"Rs."<<q*par;
      if(r>0)
         cout<<"\n\nCOCONUT HAIR
OIL"<<setw(29)<<r<<setw(28)<<"Rs."<<r*cor;
      if(s>0)
         cout<<"\n\nTITAN WATCH"<<setw(34)<<s<<setw(28)<<"Rs."<<s*twr;
      if(t>0)
         cout<<"\n\nLUX SOAP"<<setw(37)<<t<<setw(28)<<"Rs."<<t*lur;
      if(u>0)
         cout<<"\n\nTIDE"<<setw(41)<<u<<setw(28)<<"Rs."<<u*tir;
      if(v>0)
         cout<<"\n\nREEBOK SHOES"<<setw(33)<<v<<setw(28)<<"Rs."<<v*rer;
            e=m*bir;
            f=n*per;
```

```
                g=o*der;
                h=p*mur;
                k=q*par;
                l=r*cor;
                w=s*twr;
                x=t*lur;
                y=u*tir;
                z=v*rer;
                sum=e+f+g+h+k+l+w+x+y+z;
                add=m+n+o+p+q+r+s+t+u+v;

cout<<"\n==========================================================
==========================";
            cout<<"\nTOTAL:"<<setw(40)<<add<<setw(27)<<"Rs."<<sum;
            cout<<"\n\nTHANK YOU .\t\t\t     VISIT US AGAIN\n";
        getch();
        clrscr();
        goto again;
        }
        else
        {goto again;

        }


        case 2:
        cout<<"\n\n\n\n\n\t\t\t\tCANTEEN DETAILS";
        cout<<"\n\n#TIMING:\n\n 11 AM TO 02 PM \n 03 PM TO 05 PM";
        cout<<"\n\n#CANTEEN WILL CLOSED ON TUESDAY";
        cout<<"\n\n#GOODS ONCE SOLD WILL NOT BE
EXCHANGED/REPLACED";
        cout<<"\n\n#TAKE YOUR SMART CARD WITH YOU";
        cout<<"\n\n#COLLECT YOUR SMART CARD & BILL AFTER
PAYMENT";
        cout<<"\n\n#DO NOT BREAK ANY CANTEEN ITEMS";
        getch();
        clrscr();
        goto again;
        case 3:
        goto main;
        case 4:
        if(m>0||n>0||o>0||p>0||q>0||r>0||s>0||t>0||u>0||v>0)
          {cout<<"\n\t\t\tItems Avaliable in Canteen.\n\n\n";
```

```
cout<<"==========================================================
========================";
        cout<<"\nITEMS"<<setw(40)<<"QUANTITY";

cout<<"\n==========================================================
========================";
        cout<<"\n\n\nBISCUTE"<<setw(38)<<abis;
        cout<<"\n\n\nPEPSODENT"<<setw(36)<<apep;
        cout<<"\n\n\nDENIM POWDER"<<setw(33)<<aden;
        cout<<"\n\n\nMUNCH"<<setw(40)<<amun;
        cout<<"\n\n\nPARK AVENUE PERFUME"<<setw(26)<<aperk;
        cout<<"\n\n\nCOCONUT HAIR OIL"<<setw(29)<<acoc;
        cout<<"\n\n\nTITAN WATCH"<<setw(34)<<atit;
        cout<<"\n\n\nLUX SOAP"<<setw(37)<<alux;
        cout<<"\n\n\nTIDE"<<setw(41)<<atid;
        cout<<"\n\n\nREEBOK SHOES"<<setw(33)<<aree;

cout<<"\n*********************************************************
*****************";
        getch();
      goto again;
        }
      else
        {{cout<<"\n\t\t\tItems Avaliable in Canteen.\n\n\n";

cout<<"==========================================================
========================";
        cout<<"\nITEMS"<<setw(40)<<"QUANTITY";

cout<<"\n==========================================================
========================";
        cout<<"\n\n\nBISCUTE"<<setw(38)<<qubis;
        cout<<"\n\n\nPEPSODENT"<<setw(36)<<qupep;
        cout<<"\n\n\nDENIM POWDER"<<setw(33)<<quden;
        cout<<"\n\n\nMUNCH"<<setw(40)<<qumun;
        cout<<"\n\n\nPARK AVENUE PERFUME"<<setw(26)<<quperk;
        cout<<"\n\n\nCOCONUT HAIR OIL"<<setw(29)<<qucoc;
        cout<<"\n\n\nTITAN WATCH"<<setw(34)<<qutit;
        cout<<"\n\n\nLUX SOAP"<<setw(37)<<qulux;
        cout<<"\n\n\nTIDE"<<setw(41)<<qutid;
        cout<<"\n\n\nREEBOK SHOES"<<setw(33)<<quree;
```

```
cout<<"\n\n*****************************************************
********************";
          getch();
        goto again;

          }
        case 5:
        cout<<"\n\n\n\n\n\n\n\n\n\n\t\t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t\t
THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t\t
THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\
t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n
\n\n\t\t\t\t   THANKS";
        sleep(1);
        clrscr();
        cout<<"\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n
\n\n\n\n\n\t\t\t\t   THANKS\n\n\n\t\t\t\tPress any key to Exit.......";
        getch();
        exit(0);
        default:
                cout<<"Your choice is wrong";
        getch();
```

```
        goto again;
         }
//getch();
// return 0;
 }

}
```

Conclusion

The quality of the system depends on number of factors such as design, development, testing and implementation. One vital aspects of the system quality is reliability. A system is said to reliable when used it doesn't produce failure that is dangerous and costly.

I have strived to prevent occurrence of errors using different methods and techniques that include error detection, correction and tolerance (e.g. Exception handling).

The system has been designed so as to ensure maximum reliability in operation with maximum errors. I have tried to achieve maximum goal given in objective.

The system is able to achieve the objective and provide the ultimate result called"**CUSTOMER SATISFACTION".**

Bibliography

At the end, the following bibliography is in no way complete. However it is an attempt to list a few number of sources and references when we have used for the development & design of the project in this elegant form.

Title of the Book	Author/Publisher
Object-Oriented Programming with c++	Robert Lafore/Techmedia
C++: The Complete Reference	Herbert Schieldt/Tata McGraw Hill
Mastering C++	Venugopal, Rajkumar, Ravishankar / Tata McGraw Hill

C++ by EwalaGausawmiEwalaGausawmi